BUDT703_Project_0507_06_Design
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1. Brand Name:

PulseTrack

Represents the heartbeat of railways across the US, symbolizing the central role in national connectivity.

2. Mission Statement:

To enhance Amtrak's operational performance and customer satisfaction by leveraging data-driven insights to analyze ridership trends, optimize resource allocation, enhance on-time performance, and maximize the value of AGR membership programs.

3. Mission Objectives:

(1) To identify the states with the highest average procurement expenses over the past three years to help optimize budget allocation and resource planning.
WH-Question:
"What are the average procurement expenses for all states over the past three years?"

(2) To analyze ridership trends and identify states experiencing the fastest growth over the past three years to support targeted marketing and service improvements.
WH-Question:
"Which states experienced the fastest ridership growth over the past three years?"

(3) To identify underperforming routes with average On-Time Performance (OTP) below 50% over the past three years and propose operational enhancements.
WH-Question:
"Which routes had an average OTP below 50% over the past three years?"

(4) To predict AGR membership for FY24 by leveraging historical growth trends and identify states with the top 10 expected AGR growth to enhance customer loyalty programs.
WH-Question:

“What are the predicted AGR membership for all states in FY24, based on the average growth rate from FY21–FY23? ”

4. Business Terms, Facts, Attributes, and Identifiers:

Business Terms	**Facts**	**Attributes**	**Identifiers**	**Cardinalities**
Ridership	ridershipAmount	ridershipState ridershipAmount_21 ridershipAmount_22 ridershipAmount_23	ridershipId	AGR to Ridership: one to one Route to Ridership:many to many
AGR	AGRAmount	AGRState AGRAmount_21 AGRAmount_22 AGRAmount_23	AGRId	AGR to Ridership: one to one
Procurement	procurementAmount	procurementState procurementAmount_21 procurementAmount_22 procurementAmount_23	procurementId	Procurement to Ridership: one to one
On-Time Performance (OTP)	OTPAmount	OTPState OTPAmount_21 OTPAmount_22 OTPAmount_23	OTPId	Route to OTP:many to many
Route	routeAmount	routeName routeType routeStation routeHostRailroad routeAmount_21 routeAmount_22 routeAmount_23	routeId	Route to Ridership:many to many Route to OTP:many to many

5. Relations:

Procurement(**procurementId**, procurementState, procurementAmount_21, procurementAmount_22, procurementAmount_23)

Ridership(**ridershipId**, ridershipState, ridershipAmount_21, ridershipAmount_22, ridershipAmount_23, *procurementId*)

AGR(**AGRId**, AGRState, AGRAmount_21, AGRAmount_22, AGRAmount_23, *ridershipId*)

OTP(**OTPId**, OTPState, OTPAmount_21, OTPAmount_22, OTPAmount_23)

Route(**routeId**, routeName, routeType, routeAmount_21, routeAmount_22, routeAmount_23)

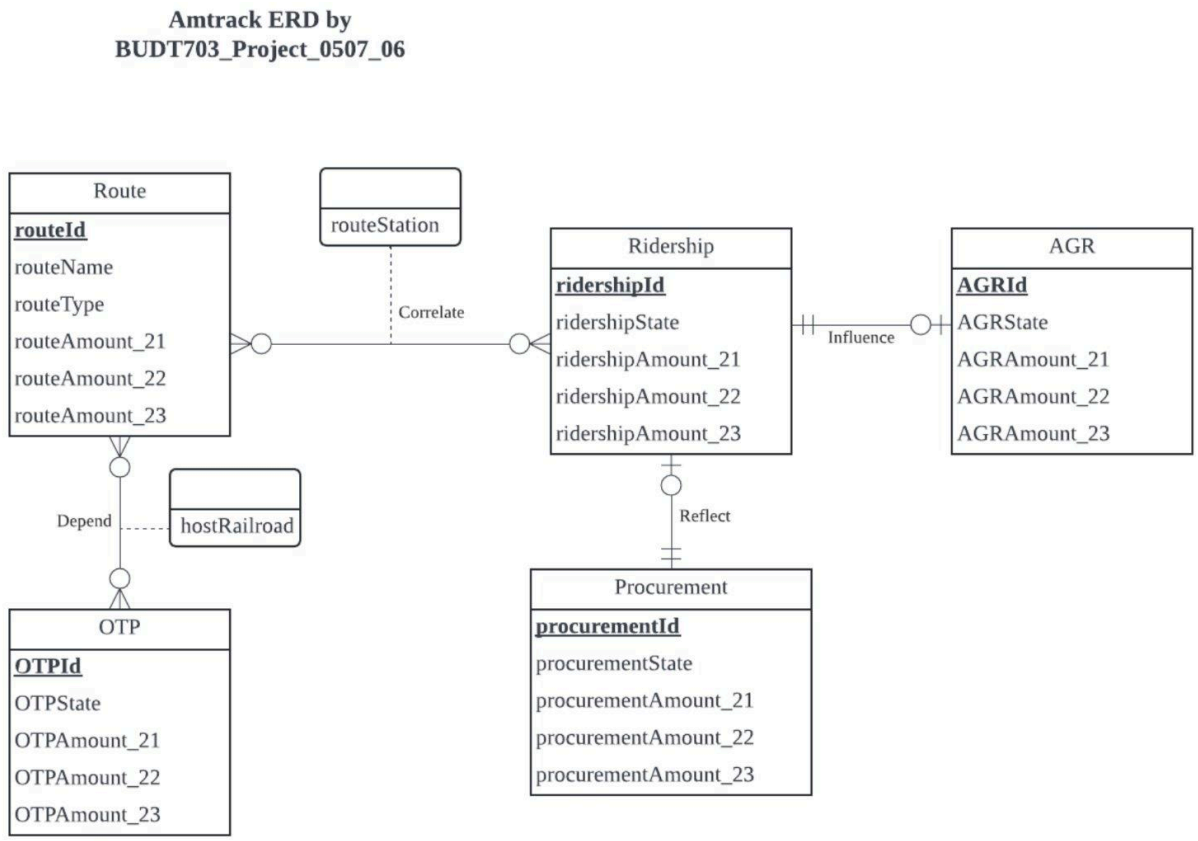
Correlate(***routeId***, ***ridershipId***, routeStation)

Depend(***routeId***,***OTPId,*** hostRailroad)

6. Referential Integrity Actions:

Relation	Foreign Key	Base Relation	Primary Key	Business Rule	ON DELETE	Business Rule	ON UPDATE
Correlate	routeId	Route	routeId	R1	CASCADE	R2	CASCADE
Correlate	ridershipId	Ridership	ridershipId	R3	CASCADE	R4	CASCADE
Depend	routeId	Route	routeId	R5	CASCADE	R6	CASCADE
Depend	OTPId	OTP	OTPId	R7	CASCADE	R8	CASCADE

7. ERD:



8. Business Rules:

[R1] When a route is deleted from the database, the other information should be deleted from the database.

[R2] When the information on a route is changed in the database, the corresponding other information should be changed accordingly.

[R3] When a ridership is deleted from the database, the other information should be deleted from the database.

[R4] When the information on a ridership is changed in the database, the corresponding other information should be changed accordingly.

[R5] When a route is deleted from the database, the other information should be deleted from the database.

[R6] When the information on a route is changed in the database, the corresponding other information should be changed accordingly.

[R7] When an OTP is deleted from the database, the other information should be deleted from the database.

[R8] When the information on an OPT is changed in the database, the corresponding other information should be changed accordingly.

9. Example Data:

Procurement
(‘P001’, ‘Alabama’, 8485192, 16401289, 30825938)

Ridership
(‘S001’, ‘Alabama’, 20603, 29086, 44170, 'P001')

AGR
(‘A001’, ‘Alabama’, 36069, 50452, 58084, ‘S001’)

OTP
(‘T004’, ‘California’, 0.587, 0.524, 0.581)

Route
(‘R001’, ‘Crescent’, ‘Long Distance’, ‘daily New York-Washington-Charlottesville-Charlotte-Atlanta-Birmingham-New Orleans’, ‘Norfolk Southern’, 0.546, 0.530, 0.570)

Correlate
(‘R001’, 'S001', 'daily New York-Washington-Charlottesville-Charlotte-Atlanta-Birmingham-New Orleans')

Depend
(‘R001’, ‘T001’, 'Norfolk Southern ')